BIMBINGAN TUGAS AKHIR #6

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1 Pembuktian Distribusi Normal $y = f(x) + \varepsilon$

Diberikan model observasi linier:

$$y = f(x) + \varepsilon = wx + \varepsilon \tag{1}$$

dengan galat $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$.

MGF Distribusi Normal:

- **Univariat**: Untuk peubah acak $Z \sim \mathcal{N}(\mu, \sigma^2)$, MGF didefinisikan sebagai:

$$M_Z(t) = \mathbb{E}\left[e^{tZ}\right] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \tag{2}$$

- **Multivariat**: Untuk vektor acak $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ dengan $\mathbf{Z} \in \mathbb{R}^k$, MGF didefinisikan sebagai:

$$M_{\mathbf{Z}}(\mathbf{t}) = \mathbb{E}\left[\exp(\mathbf{t}^T\mathbf{Z})\right] = \exp\left(\mathbf{t}^T\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}^T\Sigma\mathbf{t}\right) \tag{3}$$

Untuk galat $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$, MGF-nya adalah $M_\varepsilon(t) = \exp\left(\frac{1}{2}\sigma_n^2 t^2\right)$.

Pembuktian distribusi observasi ditinjau dalam dua kasus nilai bobot:

1. **Kasus 1**: Bobot bernilai konstan ($w \in \mathbb{R}$ atau $\mathbf{w} \in \mathbb{R}^d$).
2. **Kasus 2**: Bobot berdistribusi normal ($w \sim \mathcal{N}(0, \sigma_w^2)$ atau $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Sigma_p)$).

1.1 Kasus 1: Bobot Bernilai Konstan

Claim 1.1 (Bentuk Skalar Univariat)**.** *Jika $w \in \mathbb{R}$ berupa skalar konstan dan galat $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$, maka variabel observasi $y = wx + \varepsilon$ berdistribusi normal:*

$$y \sim \mathcal{N}\left(wx,\ \sigma_n^2\right) \tag{4}$$

Proof. MGF dari y:

$$\begin{aligned}
M_y(t) &= \mathbb{E}\left[e^{t(wx+\varepsilon)}\right] \\
&= \mathbb{E}\left[e^{twx}e^{t\varepsilon}\right] \\
&= e^{twx}\,\mathbb{E}\left[e^{t\varepsilon}\right] \\
&= e^{twx}M_\varepsilon(t) \\
&= e^{twx}\exp\left(\frac{1}{2}\sigma_n^2 t^2\right) \\
&= \exp\left((wx)t + \frac{1}{2}\sigma_n^2 t^2\right)
\end{aligned} \tag{5}$$

Berdasarkan bentuk MGF normal univariat pada Persamaan (2) dengan $\mu = wx$ dan $\sigma^2 = \sigma_n^2$, terbukti bahwa:

$$y \sim \mathcal{N}\left(wx,\ \sigma_n^2\right) \tag{6}$$

□

Claim 1.2 (Bentuk Vektor Multivariat: Persamaan (2.3) GPML)**.** *Untuk sekumpulan input* $X = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$*, bobot* $\mathbf{w} \in \mathbb{R}^d$ *bernilai konstan, dan galat* $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 I_n)$ *(Rasmussen & Williams, 2006, hlm. 9), distribusi bersyarat observasi* $\mathbf{y}$ *berdistribusi normal multivariat:*

$$\mathbf{y} \mid X, \mathbf{w} \sim \mathcal{N}\left(X^T\mathbf{w},\ \sigma_n^2 I_n\right) \tag{7}$$

dengan fungsi kepadatan probabilitas (likelihood) yang terfaktorisasi:

$$p(\mathbf{y} \mid X, \mathbf{w}) = \prod_{i=1}^{n} p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{1}{(2\pi\sigma_n^2)^{n/2}} \exp\left(-\frac{1}{2\sigma_n^2}|\mathbf{y} - X^T\mathbf{w}|^2\right) \tag{8}$$

Proof. MGF bersyarat dari $\mathbf{y} \mid X, \mathbf{w}$ didefinisikan sebagai:

$$\begin{aligned} M_{\mathbf{y}|X,\mathbf{w}}(\mathbf{t}) &= \mathbb{E}\left[\exp\left(\mathbf{t}^T(X^T\mathbf{w} + \boldsymbol{\varepsilon})\right) \mid X, \mathbf{w}\right] \\ &= \exp\left(\mathbf{t}^T(X^T\mathbf{w})\right) \cdot \mathbb{E}\left[\exp\left(\mathbf{t}^T\boldsymbol{\varepsilon}\right)\right] \\ &= \exp\left(\mathbf{t}^T(X^T\mathbf{w})\right) \cdot M_{\boldsymbol{\varepsilon}}(\mathbf{t}) \\ &= \exp\left(\mathbf{t}^T(X^T\mathbf{w}) + \frac{1}{2}\mathbf{t}^T\left(\sigma_n^2 I_n\right)\mathbf{t}\right) \end{aligned} \tag{9}$$

Berdasarkan MGF normal multivariat pada Persamaan (3) dengan $\boldsymbol{\mu} = X^T\mathbf{w}$ dan $\Sigma = \sigma_n^2 I_n$, terbukti bahwa $\mathbf{y} \mid X, \mathbf{w} \sim \mathcal{N}\left(X^T\mathbf{w},\ \sigma_n^2 I_n\right)$.

Karena $\Sigma = \sigma_n^2 I_n$ diagonal, suku eksponen pada MGF (9) dapat diuraikan:

$$M_{\mathbf{y}|X,\mathbf{w}}(\mathbf{t}) = \prod_{i=1}^{n} \exp\left(t_i(\mathbf{x}_i^T\mathbf{w}) + \frac{1}{2}\sigma_n^2 t_i^2\right) = \prod_{i=1}^{n} M_{y_i|\mathbf{x}_i,\mathbf{w}}(t_i) \tag{10}$$

Karena MGF gabungan bersyarat sama dengan hasil kali MGF masing-masing variabel, maka setiap $y_1, \ldots, y_n$ saling bebas dengan $y_i \mid \mathbf{x}_i, \mathbf{w} \sim \mathcal{N}(\mathbf{x}_i^T\mathbf{w}, \sigma_n^2)$. Oleh karena itu, fungsi kepadatan bersama terfaktorisasi menjadi:

$$p(\mathbf{y} \mid X, \mathbf{w}) = \prod_{i=1}^{n} p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{1}{(2\pi\sigma_n^2)^{n/2}} \exp\left(-\frac{1}{2\sigma_n^2}|\mathbf{y} - X^T\mathbf{w}|^2\right) \tag{11}$$

□

1.2 Kasus 2: Bobot w Berdistribusi Normal ($w \sim \mathcal{N}(0, \sigma_w^2)$)

Diberikan $w \sim \mathcal{N}(0, \sigma_w^2)$ dan diasumsikan w dan ε saling bebas.

Claim 1.3. *Fungsi* $f(x) = wx$ *berdistribusi normal:*

$$f(x) \sim \mathcal{N}\left(0,\ x^2\sigma_w^2\right) \tag{12}$$

Proof. MGF dari w adalah $M_w(t) = \exp\left(\frac{1}{2}\sigma_w^2 t^2\right)$. MGF dari $f(x) = wx$:

$$\begin{aligned} M_{f(x)}(t) &= \mathbb{E}_w\left[e^{t(wx)}\right] = \mathbb{E}_w\left[e^{(tx)w}\right] = M_w(tx) \\ &= \exp\left(\frac{1}{2}\sigma_w^2(tx)^2\right) \\ &= \exp\left(\frac{1}{2}t^2\left(x^2\sigma_w^2\right)\right) \end{aligned} \tag{13}$$

Sesuai bentuk MGF normal univariat pada Persamaan (2) dengan $\mu = 0$ dan variansi $\sigma^2 = x^2\sigma_w^2$, terbukti bahwa:

$$f(x) \sim \mathcal{N}\left(0,\ x^2\sigma_w^2\right) \tag{14}$$

□

Claim 1.4. *Variabel observasi $y = f(x) + \varepsilon$ berdistribusi normal:*

$$y \sim \mathcal{N}\left(0,\ x^2\sigma_w^2 + \sigma_n^2\right) \tag{15}$$

Proof. Karena w dan ε saling bebas, maka $f(x)$ dan ε saling bebas. MGF dari $y = f(x) + \varepsilon$:

$$\begin{aligned} M_y(t) &= \mathbb{E}_{w,\varepsilon}\left[e^{t(f(x)+\varepsilon)}\right] \\ &= \mathbb{E}\left[e^{tf(x)}\right] \cdot \mathbb{E}\left[e^{t\varepsilon}\right] \\ &= M_{f(x)}(t) \cdot M_\varepsilon(t) \\ &= \exp\left(\frac{1}{2}t^2\left(x^2\sigma_w^2\right)\right) \cdot \exp\left(\frac{1}{2}\sigma_n^2 t^2\right) \\ &= \exp\left(\frac{1}{2}t^2\left(x^2\sigma_w^2 + \sigma_n^2\right)\right) \end{aligned} \tag{16}$$

Sesuai bentuk MGF normal univariat pada Persamaan (2) dengan $\mu = 0$ dan variansi $\sigma^2 = x^2\sigma_w^2 + \sigma_n^2$, terbukti bahwa:

$$y \sim \mathcal{N}\left(0,\ x^2\sigma_w^2 + \sigma_n^2\right) \tag{17}$$

□

Claim 1.5 (Bentuk Vektor Multivariat)**.** *Untuk sekumpulan input $X = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^T \in \mathbb{R}^{n\times d}$, bobot $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Sigma_p)$, dan galat $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 I_n)$ dengan $\mathbf{w}$ dan $\boldsymbol{\varepsilon}$ saling bebas:*

$$\mathbf{f} \sim \mathcal{N}\left(\mathbf{0},\ X\Sigma_p X^T\right) \quad \textit{dan} \quad \mathbf{y} \sim \mathcal{N}\left(\mathbf{0},\ X\Sigma_p X^T + \sigma_n^2 I_n\right) \tag{18}$$

Proof. Berdasarkan MGF normal multivariat pada Persamaan (3):

$$M_{\mathbf{f}}(\mathbf{t}) = \mathbb{E}\left[\exp(\mathbf{t}^T X\mathbf{w})\right] = \exp\left(\frac{1}{2}\mathbf{t}^T\left(X\Sigma_p X^T\right)\mathbf{t}\right) \implies \mathbf{f} \sim \mathcal{N}\left(\mathbf{0},\ X\Sigma_p X^T\right) \tag{19}$$

Karena $\mathbf{w}$ dan $\boldsymbol{\varepsilon}$ saling bebas:

$$\begin{aligned} M_{\mathbf{y}}(\mathbf{t}) &= M_{\mathbf{f}}(\mathbf{t}) \cdot M_{\boldsymbol{\varepsilon}}(\mathbf{t}) \\ &= \exp\left(\frac{1}{2}\mathbf{t}^T\left(X\Sigma_p X^T + \sigma_n^2 I_n\right)\mathbf{t}\right) \implies \mathbf{y} \sim \mathcal{N}\left(\mathbf{0},\ X\Sigma_p X^T + \sigma_n^2 I_n\right) \end{aligned}$$

□

2 Transisi dari DNN Menuju DGP

Deep Neural Network (DNN):

$$\mathbf{h}^{(0)} = \mathbf{x} \in \mathbb{R}^{d_0}$$
$$\mathbf{h}^{(\ell)} = \sigma\left(W^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\right), \quad \ell = 1, \ldots, L \tag{20}$$
$$\hat{\mathbf{y}} = \mathbf{h}^{(L)} = (f_L \circ \cdots \circ f_1)(\mathbf{x}) \tag{21}$$

dengan parameter bobot deterministik $W^{(\ell)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, bias $\mathbf{b}^{(\ell)} \in \mathbb{R}^{d_\ell}$, dan fungsi aktivasi non-linier $\sigma(\cdot)$.

Gaussian Process (GP) Standar:

$$f(\mathbf{x}) \sim \mathcal{GP}\left(0,\ k(\mathbf{x}, \mathbf{x}')\right), \quad y = f(\mathbf{x}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(0,\ \sigma_n^2\right) \tag{22}$$

dengan fungsi kovariansi tunggal $k(\mathbf{x}, \mathbf{x}')$.

Deep Gaussian Process (DGP):

$$\mathbf{h}^{(0)} = \mathbf{x}$$
$$\mathbf{h}^{(\ell)} = \mathbf{f}^{(\ell)}(\mathbf{h}^{(\ell-1)}) + \boldsymbol{\varepsilon}^{(\ell)}, \quad \ell = 1, \ldots, L \tag{23}$$

di mana pemetaan berbobot $(W^{(\ell)}, \mathbf{b}^{(\ell)})$ pada (20) digantikan oleh fungsi dari distribusi proses Gaussian ($\mathcal{GP}$):

$$\mathbf{f}^{(\ell)} \sim \mathcal{GP}\left(\mathbf{0},\ K^{(\ell)}\right), \quad K_{ij}^{(\ell)} = k^{(\ell)}\left(\mathbf{h}_i^{(\ell-1)}, \mathbf{h}_j^{(\ell-1)}\right), \quad \boldsymbol{\varepsilon}^{(\ell)} \sim \mathcal{N}\left(\mathbf{0},\ \sigma_\ell^2 I\right) \tag{24}$$

Komposisi luaran total:

$$\mathbf{y} = \left(\mathbf{f}^{(L)} \circ \cdots \circ \mathbf{f}^{(1)}\right)(\mathbf{x}) + \boldsymbol{\varepsilon} = \mathbf{f}^{(L)}\left(\ldots \mathbf{f}^{(1)}(\mathbf{x}) \ldots\right) + \boldsymbol{\varepsilon} \tag{25}$$